

Money, Credit and Banking

Money is ubiquitous in society but the definition of money is still perennially being debated by economists and anthropologists. As this is a finance textbook, a somewhat unconventional approach will be taken here. Starting from abstract properties of money, the concrete items that can be used as money will be explained in a roughly historical order. Starting from the abstract definition has the advantage of being able to focus on essential features before getting side-tracked by ancillary properties.

2.1 ABSTRACT PROPERTIES OF MONEY

Money is a tool that enables people to trade goods and services against a standard *means of exchange*. This function makes money an abstract commodity in the sense that is received and given as payment for all other goods or services without necessarily having a material value in itself¹. Just what constitutes money in this sense in a given society is decided by common agreement because everybody involved in trading needs to accept a particular form of money for it to be effective. There does not have to be a unique money in a given society. Multiple commodities can serve as money in parallel and most modern monetary systems use at least two forms of money.

When transactions involve money on one side, it makes sense to express the value of assets in terms of the amount of money they can be exchanged against. This leads to the use of money as a *unit of account*.

A concomitant requirement of a means of exchange is that it holds its value, at least over short periods. This requirement means that the rate at which money can be exchanged into goods and services does not fluctuate too much from one day to the next. Sellers of goods or services against money need to have confidence that the money received will purchase a known quantity of other services in the near future. Money can therefore serve as a *store of wealth*.

The use of money is therefore intrinsically linked to the notion of credit: By accepting money (instead of goods or services) for payment, a seller of goods defers consumption in a way that would not be automatic in barter trade. Money can in this sense be

¹This definition does not rule out that money can also be a concrete commodity with an intrinsic value.

thought of as an obligation of society as a whole to provide goods and services to holders of currency².

The credit aspect of money implies that money is not in itself wealth but merely a claim on actual goods and services. A society can choose to increase the amount of money in circulation but will not usually find itself richer as a result. Most successful instances of money creation for the purpose of economic stimulus can be traced to increases of credit, i.e., an increase in economic activity not tied to current expenditure. However, the creation of excessive amounts of currency (fiat or specie) can lead to lower economic activity. For instance, inflow of specie from the Americas was argued to have depressed economic activity in the countries that directly benefited from these inflows (Portugal and Spain) (cf. Cantillon in [60]).

Money is therefore a social construct creating an asset that can be used in exchange, as a unit of account, and as a store of value. Among the assets that serve as money, there is generally a subset known as cash that has crucial specific properties that are sometimes overlooked by classical economists:

Immediate finality of payment Cash transfers are non-reversible and therefore final as soon as they occur unless there is an obvious evidence of theft. In contrast, even simple non-cash transactions such as credit card payments are subject to potential unwind risks related to uncovered obligations.

Anonymity Two parties can transact in cash without knowing the identity of the other party.

Decentralisation Cash transactions do not rely on third parties, in particularly not on a bank providing account services.

Indentifiability Users should be able to readily identify the asset and distinguish it from counterfeits.

Cash transactions incur significant costs related to the safe storage, distribution, collection and counting of cash. Because cash can be transferred quickly and anonymously, there is much more of an incentive to steal cash than any other asset. Stocking cash machines in a remote location of a large country, and returning cash taken in the shops of that location to the central bank requires a lot of transportation in special trucks. Users of the cash usually do not consider such costs which are, however, intimately related to these four properties.

²Some economists assume instead that money is an obligation of the central government under a theory called chartalism or, more optimistically, Modern Monetary Theory (MMT). Such thinking might be encouraged by evidence like Section 16 of the Federal Reserve Act [34] which states that bank notes ‘shall be obligations of the United States’. The motivation for this statement is likely to have been that in 1913 when the Federal Reserve System was set up bank notes were issued by private banks which could be subject to defaults. By explicitly putting the credibility of the federal government behind the ‘greenback’, the stability of the new paper currency was enhanced. This does not negate the self-referential nature of fiat money. The link between government debt and currency is contradicted not only by cases where currency is used even though the central government has no credibility (the Somali Shilling at the time of writing), but also cases where societies use money issued in other jurisdictions, i.e., cases of dollarisation.

2.2 EARLY FORMS OF MONEY

The oldest form of money is physical and many still think of money in physical terms. As long as there is sufficient stability in the supply of any commodity and its value in terms of other commodities can therefore be reliably assessed, it can be used as money. Money that is based on the value of the commodity of which it is made is known as commodity money. In the larger Asian and Western societies, money used to be associated with precious metals, namely gold and, to a smaller degree, silver. It is therefore common to think of money as something that has value in and of itself, leading to the idea that the 'gold standard', gold-based money, is something inherently desirable. Other cultures used standard commodities, such as rice or grain, to express and settle obligations. However, the exclusive use of commodities as money is neither the historical origin of money, nor is it its apparent direction. Before discussing the basics of modern monies for the purposes of trading in the money market, it is instructive to spend some time on the history of money.

Seen as a tool, cash should be reasonably portable, easily recognisable and difficult to counterfeit. In central Europe, where mining was introduced fairly late, commodities such as salt played the role of money for a long time. Salt was sufficiently scarce to give small amounts of it appreciable value; it can be kept for a long time; and it is difficult to forge. Salt is also an essential food ingredient, so salt obtained in trade could be consumed if it wasn't needed for as a store of wealth. This is only one example for using consumables as money. Islamic law recognises six commodities that must be traded 'equal for equal, and hand to hand' (gold, silver, wheat, barley, dates and salt). Islamic law referring to these six commodities is deemed to apply to money in general and therefore governs, for instance FX transactions.

When an item used as money has an idiosyncratic value, for instance under the gold standard, the credit element of money can be obscured. Receivers of money then receive a valuable asset as well as an abstract commodity. However, the production value of gold, or any other form of commodity money, can fluctuate relative to that of other commodities, for instance as a result of advances in mining technology. The historical stability of a commodity as a means of payment is therefore probably at least in part due to its function of money. Users of commodity monies accept the commodity with a view to its future value for purchases, i.e., its monetary value, rather than for its intrinsic value alone. The validity of this argument is difficult to prove conclusively with historical data. It amounts to the statement that stability of prices under a gold standard is not just the result of the value of gold anchoring the value of other goods, but that conversely the value of gold is also anchored by the value of other goods. In the current, post-metallic currency system, the predominant role of the US dollar as an international trade invoice currency is likely to stabilise its exchange value³.

The volatility of gold prices since the demonetisation of gold in the second half of the 20th century is an illustration, albeit an imperfect one, of this point. Figure 2.1 shows the ratio of the gold price in dollars to the urban consumer price index, which

³This is the essence of the 'exorbitant privilege' enjoyed by the US, although the deep liquidity of the dollar-denominated capital markets also support the external value of the dollar.

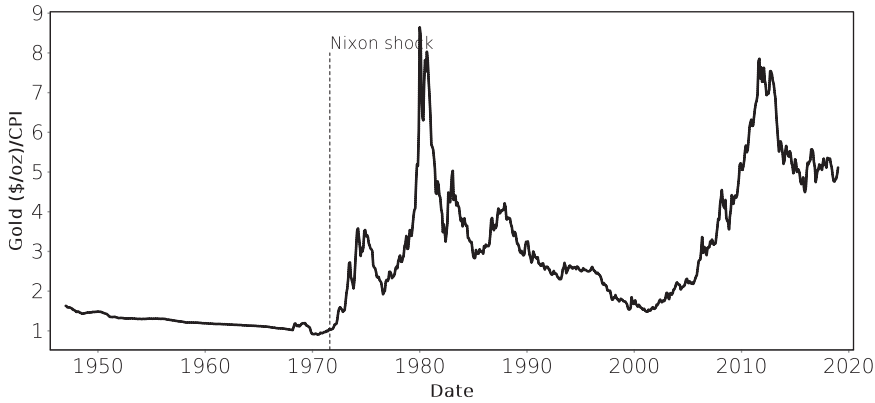


FIGURE 2.1 Gold price (London fixing) in US dollars divided by the US Urban Consumer Price index. Data retrieved from FRED.

is a measure of the price of a standard basket of goods and services in the US. A stable exchange ratio between gold and goods and services would imply a low volatility of this series. The data appears to bear out the idea that monetary use of gold may have stabilised this ratio before the suspension of dollar convertibility into gold by President Nixon on 15 August 1971 made it more volatile.

However, the technical background is more complex. Under the Bretton Woods system, the dollar was pegged to gold which required the active control of gold supply and demand by central banks. The dollar prices of other goods and services meanwhile followed other economic trends, including the policies of the Federal Reserve System. Stability of gold relative to goods and services in the US was therefore due to a common anchor, namely the US dollar. The higher volatility of gold relative to other prices following the suspension of gold convertibility is therefore not only due to changes in market mechanisms but the result of changed government policies. At the same time, gold sometimes outperforms other assets in times of crisis because some investors expect it to become more valuable as a monetary commodity.

Metals, especially precious metals, make the best commodity monies because they are easy to store, count, and transfer. However, at most times in history, commodity monies in practice have traded at a premium or discount to the actual commodity. If money can be made from the metal and turned back into the non-monetary metal itself (viz. the minting and melting of coins), the cost of minting or melting provides a corridor around the inherent metal value within which the commodity money can trade. Normally, minted money is slightly more expensive than its commodity value because it is more convenient to use as a means of payment. In some cases, minting is restricted to certain workshops operated or licensed by a sovereign and money creation is subject to a special form of tax, known as seigniorage [71]. The value V of a coin is then given by the following inequality involving the inherent metal value V_i , minting cost V_M , melting cost V_m and seigniorage S :

$$V_i - V_m \leq V \leq V_i + V_M + S \quad (2.1)$$

If a coin is trading below the lower of these boundaries, scrap metal merchants would find it attractive to melt it down, while if coins are trading above the boundary, it would

be attractive to mint new coins. The use of coins introduces a number of interesting economic problems, but these are of no interest for the financial markets. The interested reader is referred to the literature, in particular [71].

A legal concept called *lex monetae* stipulates that each sovereign can legislate what constitutes money in its own jurisdiction. This concept is at the same time trivial and misleading. A sovereign can certainly legislate what types of assets can be used to discharge debts stated in monetary terms, at least as far as court-enforced payments are concerned. Such assets are referred to as legal tender or forced money. Given that sovereigns are usually able to levy taxes, and therefore command large payment flows, prescribing legal tender can lend significance to assets by legal fiat. While *lex monetae* in this sense is not written legislation, its operation can be found already in old international treaties. The Pax Westphalica, the peace treaty that in 1648 ended the hostilities of the 30 Years War and set important precedents for international law in Europe, specified restitution payments in the local currency of the respective sovereigns [67]. That being said, those sovereigns were all using gold as the base of their currencies at the time. On this evidence, *lex monetae* amounts to the acknowledging that sovereigns have the right to stamp their own coinage.

Beyond this trivial scope of *lex monetae*, societies can find themselves in situations where the legislated money is unsuitable for daily transactions and choose to use other means of exchange, such as money issued in other jurisdictions. The misleading aspect of *lex monetae* therefore is that it can suggest that full control of a sovereign over the money used in its jurisdiction is at all times assured. Taken to the extreme, it would seem to extend to giving sovereigns the right to change the value of monetary claims by changing the national currency⁴. While possible in some national laws, the absence of *lex monetae* in international treaties means that this option is unavailable in cross-border funding.

Depending on the money used, the actual inherent metal value was sometimes difficult to estimate because coins with different degrees of purity were in circulation, or the weight of a coin could not easily be assessed. When a government deliberately brings coins of lower purity into circulation this is called debasement of a currency⁵. Various techniques to achieve the same effect were invented by private forgers. For instance, until Sir Isaac Newton invented the rolling of coins, the practice of cutting off the edges of coins (clipping) was used to reduce the weight of coins relative to their nominal value.

In other parts of the world, island tribes in the Carribean used sea shells or curiously shaped stones as money, known as wampun. The conceptual importance of wampun

⁴During the euro crisis in 2010, some commentators suggested that *lex monetae* would give euro area member states the right to repay euro-denominated debts in devaluated domestic currencies without being in default.

⁵It is the history of debasements conducted in order to reduce government debt that excited the aficionados of *lex monetae* at the height of the euro crisis. Reverting to a national currency which would then be devalued against the euro was seen as a way to reduce excessive debt burdens in some euro members states. The fundamental flaw with this argument, as far as EU Law is concerned, is that article 118 of the Treaty on the Functioning of the European Union states that the euro is the currency of the European Union. Whatever rights *lex monetae* may be conferring, euro area member states have signed them away. In addition, changing the denomination of an existing security to the detriment of bond holders is likely to be an illegal unilateral contract modification in most legal systems.

relative to the use of precious metal is that wampun has no inherent commodity value at all. It is therefore money that derives its value purely by social agreement. Money that is defined purely through social agreement in this way is known in economics as fiat money⁶. All modern currencies are fiat monies in this sense. However, the discussion about commodity money above shows that even commodity monies usually have a value component that is no different from fiat money. In particular, seigniorage introduces such a component into a commodity money. The act of minting is only important if there is a social convention to use particular coins, or a particular advantage in doing so.

2.2.1 Paper money and bank notes

Transacting large amounts of trading using physical coins is difficult and between the 9th and 11th centuries, the Chinese monetary system introduced paper money to make trading easier. In addition, frequent episodes of debasement eroded the faith in the coins of some jurisdictions. This created an incentive for merchants to establish payment instruments linked to metal of a known purity and weight. When paper money is discussed in the context of commodity money, the underlying commodity is usually referred to as specie.

Two very different forms of paper money exist, namely certificates and bank notes. A certificate is documentary evidence of a particular amount of metal (or other commodity money) that is stored somewhere in physical form. Certificates are therefore simply a more convenient way of transacting trades in commodity money and this is the original form of paper money used in China. Certificates are always fully backed by the physical commodity and holders of the certificate can easily exchange the paper for the physical commodity.

A bank note in the current sense, however, is merely a document stating the promise of a particular issuing bank to pay a certain amount of money. In other words, bank notes are debt certificates that are payable on demand. Among the major currencies, the English pound is one of the few to still retain this notion and pound notes state this payment promise explicitly, as shown in Figure 2.2.

Bank notes in Europe developed naturally from certificate monies. The point at which a depositary receipt becomes a bank note is when the issuer of the note can lend the physical money to a third party, i.e., when the trustee becomes a banker. The transition from the sole use of specie or certificates for payment towards bank notes represents an important transition in the definition of money: When paper money is accepted as payment in transactions, the banks issuing this paper create money. As a result, two parallel monies exist in the economy, namely the original specie (outside money) and the paper money and deposit money issued by banks (inside money). These two monies are in principle interchangeable, at least as long as the public has no doubts as to the ability of banks to pay out specie on demand.

⁶Fiat is latin for 'let there be'.



FIGURE 2.2 Detail of an old five pounds sterling bank note. Because this note was legal tender in England, the Bank of England could keep the promise by returning the note to the bearer. Source: Bank of England.

2.3 FIAT MONEY

The last step in the creation of modern money is to realise that the deposit of specie does not actually need to exist as long as some form of bank note is generally accepted as a means of payment. Indeed, as Kynaston pointed out [53], the suspension of cash (gold) payment on Bank of England notes in 1797 marked the transition to paper money, and this transition was made possible by concomitant declarations by merchants that they would accept the paper notes as payments despite their suspended convertibility into specie⁷.

To stay with the English pound as an example; the promise shown in Figure 2.2 is economically meaningless because anybody delivering this note to the Bank of England with the demand to repayment could in practice be given the same note back. The note is a legal instrument for discharging debts in England (legal tender) and therefore the obligation of the Bank would be fulfilled by the return of the note. For this reason, the US dollar notes simply state that they are legal tender, and the notes of most other currencies, including euros and yen, simply state a nominal amount. Through the circuitous route of metal-based money, modern economies now have established a cash system that is based purely on social contract. The use of token monies in this system is similar to wampun.

2.3.1 Fiat money and trade

For a metal-based currency to work, there must be enough physical metal available to withstand a bank run. Banks in a metallic system generally hold a so-called reserve of

⁷Technically speaking, 1797 was not the first time the Bank of England suspended payments in specie because such an episode occurred already more than a hundred years earlier. What made 1797 special is that the suspension was mandated by law. Proponents of the gold standard would do well to remember that such currency arrangements tend to be suspended just when they are supposed to be most attractive, namely in times of crisis. In 1797, the immediate cause of the crisis was the onset of the Napoleonic Wars.

specie that is designed to cover redemption requests, as well as specie required for inter-bank transactions. This reserve can be held individually or through a central reserve bank, but must be sufficient to satisfy the expected maximum amount of notes that could be presented for payment. The ratio of reserves held to the amount of banknotes outstanding is called the reserve ratio.

This requirement creates a straightjacket for money creation because the reserve ratio, and hence the monetary multiplier in Equation (4.3) cannot easily be altered. In effect, the amount of money that can be created in such an economy is limited by what that economy can source in terms of reserve metal. For this reason, some economists advocate the return to a metal-based monetary system because they see pure fiat currencies as an easy temptation for inflationary policies (see e.g. Friedman [37] for an entertaining discussion). Indeed, when sovereigns have replaced metal-based currencies with fiat money, the intention was often to raise funds at the expense of the general population through deliberate debasement, leading to bouts of high inflation. In particular, episodes of paper money in economies that were otherwise using metal-based currencies are usually associated with wars.

However that may be, the mathematical realities of modern economies make the idea of meaningful commodity links unrealistic in any case. The 'straightjacket' mentioned above has become too tight for the world economy. At the time of writing, the quarterly trade deficit of the United States of America alone is in the same order of magnitude as the annual world gold production⁸. In international commerce, where national laws regarding bank reserves are largely irrelevant, this trade deficit would have to be covered with large outflows of physical gold⁹. Even if the monetary gold holdings of the USA were able to cover this outflow for a sustained period, the loss of reserve metal would lead to an eventual contraction of the monetary base in the USA¹⁰.

By instead settling international trade in fiat money, trade volumes are not constrained by the availability of a specific physical commodity. The physical constraint is replaced by one based on expectations. A persistent large trade deficit creates doubts about the value of holding the currency of that country. This will over time lead to a cheapening of that currency, hopefully making the country more competitive in international trade, and correcting the trade imbalance. Whether either of these two adjustments works in practice is a matter of conjecture because every exchange rate involves two currencies, and cross-border fund flows can also be affected by investment in financial instruments and real assets.

⁸The US trade deficit in 2019 averaged USD 51.4bn per month while the 2018 world gold production of 3,332 tons had a market value of around USD 135.5bn using the average London fixing (Source: World Gold Council, FRED).

⁹Persistently large gold outflows were the driver for the suspension of dollar gold convertibility by President Nixon on 15 August 1971.

¹⁰The opium wars between the UK and China are often linked to the drain of sterling silver from the UK to India which had to be offset by silver income from selling drugs to China.